

MATLAB simulation codes for [1, Section 4]

1. Overview

This MATLAB code simulates pedestrian (crowd) dynamics and infection propagation in a confined domain with a single exit using a second-order macroscopic model from [1, Sect 4].

It couples:

- A **hydrodynamic model** for pedestrian density and velocity (Roe's scheme with MUSCL reconstruction);
- A **drift–diffusion–reaction model** for airborne infection potential (Rusanov flux);
- A **SEIS compartmental model** for infection spread (susceptible–exposed–infected–protected).

The simulation outputs real-time contour maps and a video showing the evolution of:

1. Total pedestrian density ρ ,
2. Infection coefficient β_I ,
3. Exposed population density ρ^E .

2. Key Numerical Methods

- **Crowd Flow Solver:** Roe's approximate Riemann solver for the second-order macroscopic flow equations [3].
- **Flux Reconstruction:** Monotone Upstream-Centered Scheme for Conservation Laws (MUSCL) with flux limiter function [4].
- **Eikonal Equation:** Solved via the Fast Sweeping Method (FSM) from [2], defining pedestrian desired directions.
- **Infection Potential β :** Solved by Rusanov's scheme with MUSCL reconstruction for the advection part and second-order finite differences for the diffusion part.
- **Epidemic Dynamics:** SEIS model coupled with pedestrian advection.

3. File Structure

Main_Crowd_SEIS.m	Main driver file
Approx_Roe_PW.m	Roe solver for flux computation
lim.m	Flux limiter function
FSG.m	Fast sweeping solver for the Eikonal equation
UV.m	Desired speed model based on local density
potential.m	For ventilation potential solver for airflow
RESULTS.m	Text file with time, total pedestrians, and exposed counts
Test_Case.mp4	Output movie showing crowd and infection evolution

4. Simulation Setup Example Problem

- **Domain:** 10 m $\times$ 10 m rectangular room.
- **Exit:** 2 m wide, centered on the right boundary.
- **Grid spacing:** $\Delta x = \Delta y = 0.1$ m.
- **Time step:** $\Delta t = 0.01$ s.
- **Initial Infected:** $\rho_0^I = 0.25\rho_0$ and $\rho_0^V = 0$
- **Total time:** 3000 iterations (≈ 30 s simulation).

Initial pedestrian density is prescribed in a rectangular region $1 \leq x \leq 5$, $2.5 \leq y \leq 7.5$ with $\rho_0 = 2.4$ pedestrians/m².

5. Model Parameters

Symbol	Meaning	Default Value
τ	Relaxation time	0.6 s
v_f	Free flow speed	1.4 m/s
R_m	Maximum density	6 ped/m ²
S	Diffusion coefficient (infection)	1.2×10^{-3}
i_0	Infection rate	0.04
ϵ_1	Numerical threshold	10^{-4}
lose	Particle loss coefficient	0.5
kap	MUSCL reconstruction parameter	0

6. Output Files

- **Test_Case.mp4:** Movie file showing temporal evolution of crowd density, infection coefficient, and exposed density.
- **RESULTS.m:** Text file containing:

time (s) total exposed pedestrians total pedestrians in domain.

7. Running the Simulation(s)

1. Ensure all required MATLAB functions (**Approx_Roe_PW.m**, **FSG.m**, etc.) are in the same directory.
2. (Optional) If including ventilation, run:

`potential`

and comment out the first line in **Main_Crowd_SEIS.m** that clears the workspace as well as `ug=zeros(Ny,Nx); vg=zeros(Ny,Nx)` in lines 390-391.

3. Run the main code:

`Main_Crowd_SEIS`

4. View progress in MATLAB figures and the generated movie.

8. Notes and Recommendations

- Adjust Δt , Δx , and τ to satisfy the Courant–Friedrichs–Lewy (CFL) condition to satisfy eq. (29) in [1]
- Modify the initial infection rate i_0 or diffusion coefficient S to simulate various contagion scenarios.
- For ventilation studies, activate airflow fields computed by `potential.m`.
- Can change initial condition of assumed infected pedestrians ρ_0^I and vaccinated/masked ones ρ_0^V
- Visualization parameters (color limits, view angles) can be modified in the plotting section.

9. References

1. A.I. Delis, N. Bekiaris-Liberis, “Numerical investigation of the effect of macro control measures on epidemic transport via a coupled PDE crowd flow - epidemic spreading dynamics model”, *Applied Mathematics and Computation*, Volume 513, 2026.
2. E. Zhao, “Fast sweeping methods for Eikonal equations,” *Math. Comput.*, 2005.
3. Roe, P. L., “Approximate Riemann solvers, parameter vectors, and difference schemes”, *J. Comput. Phys.*, 1981.
4. E.F. Toro, *Riemann Solvers and Numerical Methods for Fluid Dynamics*, Springer Berlin, Heidelberg, Heidelberg, 2009.